

K1356 — DAY CLOSE 2026-08-10. The $\sin^2\theta_W$ honest disposition (the scale gate cut against the pretty answer, as it should): NO Weinberg win. The “real-dim doubling $\rightarrow 3/13$ ” lead is REFUTED (blind $\kappa=5/3$; and $3/13$ -at- μ_{geo} runs away from $3/13$). The inner-fluctuation route UNDERSHOOTS ($3/8$ from near-Planck $\rightarrow \sim 0.20$, missing measured 0.231). The ONLY viable route to the measured angle is the scale-free conformal-boundary curvature grading ($3/13$, F531) — and its scale is UNPINNED. That is tomorrow’s ballgame. + the day’s completed arc + tomorrow’s queue.

Keeper (day close, 2026-08-10 ~17:45 EDT. Casey called rest. The scale gate (K1354/K1355) I raised did its job — it caught the day’s most seductive over-reach, and Cal/Grace/Lyra/Elie all owned it flat and computed the honest run-down. The discipline was the product. Recording the disposition, what’s complete, and tomorrow’s queue. Cal cold-reads tomorrow.)

★ THE $\sin^2\theta_W$ DISPOSITION (honest, no win — the method working)

- The “doubling $\rightarrow 3/13$ ” lead is REFUTED. Cal’s blind computation gives $\kappa = 5/3$ (the standard/complex normalization $\rightarrow 3/8$), not $10/3$. And even granting $\kappa=10/3$, $3/13$ as a μ_{geo} value runs AWAY from $3/13$ at M_Z — a boundary value can’t sit at the measured angle.
- The inner-fluctuation route UNDERSHOOTS. BST’s cutoff $\mu_{geo} \approx 1.6 \times 10^{18}$ GeV (near-Planck, from $\ell_B \approx 7.82 \ell_P$). Running $3/8$ down from there with the SM RGE lands at ≈ 0.20 , missing the measured 0.23122 by ~ 0.03 (the well-known SM non-unification, made *worse* by the near-Planck start — exactly the K1355 quantitative hurdle, now computed). So “ $3/8$ runs to the measured angle” is FALSE without new physics, and BST is NOT the failed-GUT story.
- The ONLY route that can match the data: the scale-free conformal-boundary curvature grading ($3/13 = N_c/(N_c+2n_C)$, #85; charge as a pure count on the conformal boundary, F531) — IF it is genuinely a low-energy quantity with nothing to run. But its scale is UNPINNED — Lyra’s own #85 memory had $3/13$ “at the geometric scale, deviations = running,” which would put it HIGH, contradicting “ $3/13$ at M_Z .” The scale of BST’s $3/13$ is the whole ballgame, and it is unresolved.
- VERDICT: NO Weinberg-angle win. $3/13$ stays a pretty fraction until its scale is pinned. The scale gate caught a genuine over-reach (Lyra/Grace/Elie all conflated a near-Planck value with a lab value because $3/13$ was the target). This is the method succeeding, not a failure — we caught it before a referee would.
- The uniquely-ours thread survives (for tomorrow): IF BST has a coupling boundary condition, the flow that carries it down is BST’s OWN induced β -function (a_2 , #17) — Connes imports the RGE. Owning both the geometry and the flow is the real game; but it only matters once the scale of $3/13$ is settled.

☐ COMPLETE and DEFENSIBLE (the day’s real, banked arc — tiers honest)

1. BST is a fully-exhibited, referee-safe real spectral triple — self-adjoint (Parthasarathy, $D^2 \geq 0$ on unitary H^2), compact resolvent, finite summability; one homogeneous Dirac; floor-shift $+6=C_2$. [T2549; Cal-ratified.] Theorem-level.
2. KO-dimension 2, primary-verified (van Suijlekom/Connes–Marcolli); the whole divergence from Connes’ SM triple is ONE sign, $\varepsilon=J^2=-1$, and it is CP. [T2550.] Derived-structure, primary-verified.
3. The finite algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ is forced (endomorphism algebra of the finite space). [T2551.] Derived-structure.
4. BST is the Standard-Model triple, not Pati–Salam — the colorless, ν_R -free neutrino makes the mixing color-diagonal $\rightarrow$ first-order holds (Elie’s computed commutator = the Five-Absences root, count once). [T2552.] Derived-structure, provisional.
5. The hypercharges reproduce the SM values (rep + unimodularity, anomaly-free). [Elie 5168.] Derived-structure.

6. Strong sector unified (UV-IR one-operator bridge; mass gap = 6th face of one spectrum); antiuniverse dynamically excluded (semigroup theorem-grade; cosmological = analogy) [T2553]; muon = Wallach edge (why-hard, not the mass) [T2554].
- CLAIM CEILING (unchanged, K1351): CAN say “a genuine real spectral triple at KO-2, distinct-by-CP-one-sign, whose chiral no- ν_R content selects the SM branch, with SM hypercharges.” CANNOT say “derives the SM” (a_4 not started) or anything about the Weinberg angle (no win — the naive routes are refuted, the scale is open). Nothing external until Cal-vet + Casey GO. CP existence-only.

☐ TOMORROW’S QUEUE (queued, no team prompt tonight per Casey — prompt in the morning)

1. ★ THE $\sin^2\theta_W$ SCALE QUESTION (the ballgame) — Lyra + Grace + Elie. Pin the scale of BST’s 3/13. Is it a genuinely scale-free, low-energy quantity from the conformal boundary (F531 — charge as a pure count; the conformal SO(4,2) boundary has no intrinsic scale), OR is it a geometric-scale value that must run (Lyra’s #85 memory)? This decides whether BST has ANY Weinberg-angle prediction. If scale-free $\rightarrow$ 3/13 is a real low-energy fact (no GUT, no SUSY, no running — Five-Absence-consistent), a genuine result. If it runs $\rightarrow$ BST has no clean Weinberg prediction and we say so. Corpus-first: F531 (conformal boundary), #85 (curvature grading), #17 (induced β). Cal cold-reads the scale claim; blind where possible; target-innocent (don’t reason toward 3/13).
2. a_4 = the full SM Lagrangian — the summit (Higgs as inner fluctuation, Yukawas), on the banked $a_0=\Lambda$, $a_2=\text{gravity}$. The bulk of “build the SM from D_IV⁵.”
3. The ν_L Majorana $\Delta L=2$ geometric source (K1347) — no ν_R , no type-I see-saw; where does the neutrino mass come from? Substrate/geometric.
4. Open, lower priority: CFS causal-action credential (B1); the crown jewel C4 (spectral = causal action, last).
 - Standing carried: lead with the operator; overdetermined $\neq$ independent votes; predict at μ_{geo} , run DOWN with measured/induced RGE, never match fractions across scales; map-before-marry; nothing external until Cal-vet + Keeper-PASS + Casey GO; CP existence-only; nothing pushed.

— Keeper, K1356, DAY CLOSE 2026-08-10. $\sin^2\theta_W$: NO WIN. Doubling $\rightarrow$ 3/13 REFUTED (blind $\kappa=5/3$; 3/13-at- μ_{geo} runs away). Inner-fluctuation route UNDERSHOOTS (3/8 from near-Planck $\mu_{\text{geo}}\approx 1.6\times 10^{18} \rightarrow \sim 0.20$, misses measured 0.231 — the K1355 hurdle computed). Only viable route = scale-free conformal-boundary 3/13 (F531/#85), scale UNPINNED = tomorrow’s ballgame. Scale gate caught the over-reach (method WIN). COMPLETE: referee-safe spectral triple (T2549), KO-2 distinct-by-CP (T2550), forced algebra (T2551), SM-not-Pati-Salam (T2552), SM hypercharges, strong sector unified, antiuniverse excluded, muon=Wallach edge. Claim ceiling unchanged (no Weinberg claim). TOMORROW: pin the scale of 3/13 (F531 conformal boundary — decides if BST has a Weinberg prediction at all); then a_4 =full SM Lagrangian; ν_L $\Delta L=2$ source. Cal cold-reads. Nothing pushed. EOD.